GETTING STARTED · IOS · IPADOS · MACOS

RadiX Dose v2.4

Build an example clinic & run your first calculation.

This guide walks you through RadiX Dose from an empty install: first you'll **build a small example clinic** (a facility, a machine, a chamber and a beam), then you'll **run a reference-dosimetry calculation** with it. The example data used here is entirely fictional — use it to learn the workflow, then replace it with your own.

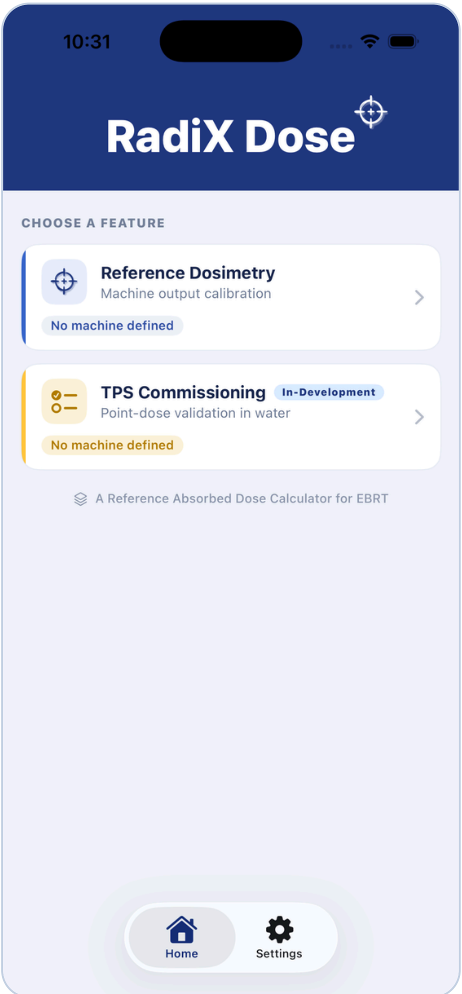
A note on the data

"General Hospital / Radiation Oncology" is a neutral example only. Never store patient-identifiable information in the app — it is built for quality-assurance data.

01 Open the app

1. **Launch RadiX Dose and accept the agreement.**
   On first launch the **User Agreement & Disclaimer** appears. Read it, tick the box, and tap **Accept** (this is asked once).

2. **Choose a feature.**
   On the home screen, tap **Reference Dosimetry** (“Machine output calibration”). The second feature, *TPS Commissioning*, is a preview and not needed here.



The home screen — tap Reference Dosimetry to begin.

02 Build the example clinic

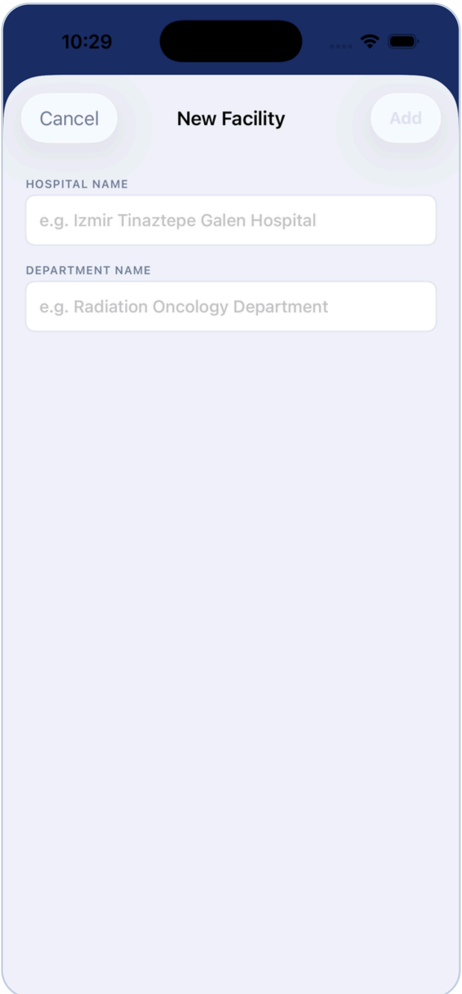
Everything a calculation needs is defined once, in the **Settings** tab (bottom-right), as a nested library: **Facility → Machine → Detectors · Electrometers · Beams**. One facility and one machine are *active* at a time.

1 Add a facility.

Go to **Settings**. Under **FACILITY**, tap **+ Add**, then fill in:

- **Hospital Name** — e.g. `General Hospital`
- **Department Name** — e.g. `Radiation Oncology`

Tap **Add**. The facility appears and becomes **ACTIVE**.



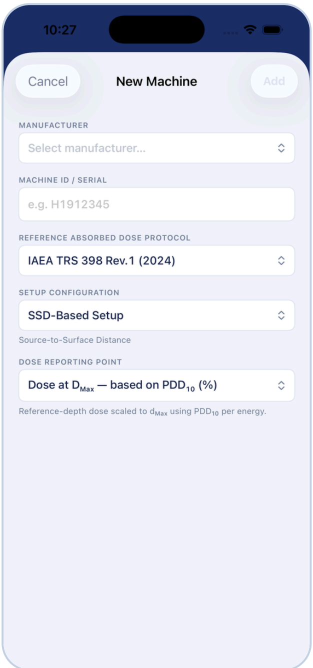
New Facility — hospital and department names.

2 Add a machine (linac).

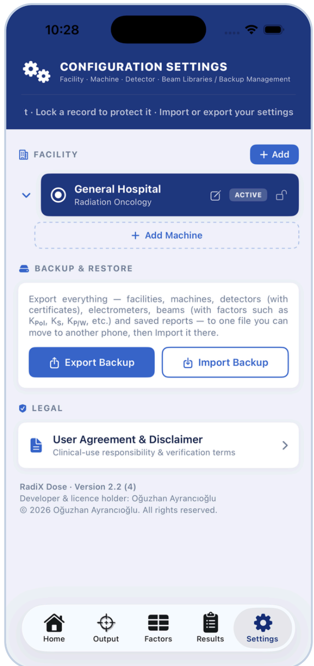
Tap the **chevron** to the left of your facility to expand it, then tap **+ Add Machine** and set:

- **Manufacturer** & **Model** — e.g. Varian · TrueBeam
- **Machine ID / Serial** — e.g. `LINAC-1`
- **Reference Absorbed Dose Protocol** — `IAEA TRS-398 Rev.1 (2024)`, `IAEA TRS-398 (2006)`, or `AAPM TG-51`. This chooses which beam-quality index and k_Q data the app uses.
- **Setup Configuration** — SSD- or SAD-based
- **Dose Reporting Point** — report at d_{max} / z_{ref}, or at the reference (calibration) depth

Tap **Add**. The machine becomes active.



New Machine — pick the code of practice.



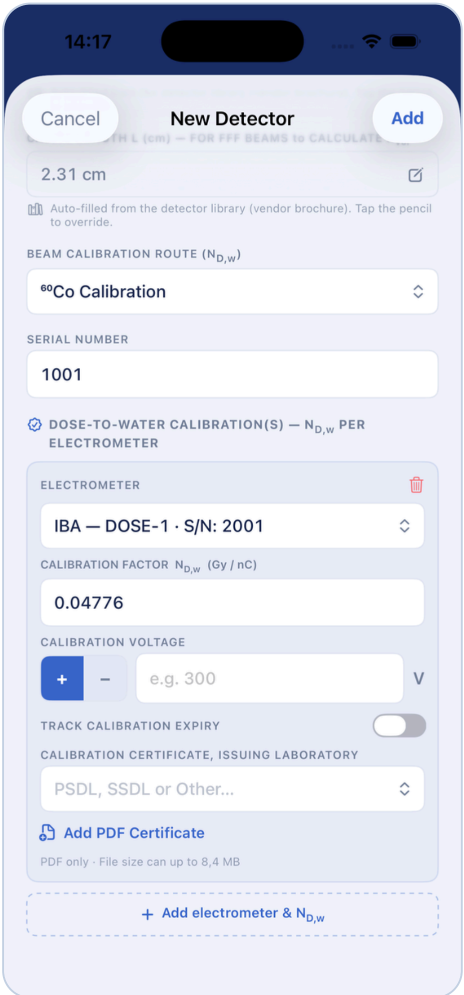
The expanded facility with “Add Machine”.

3 Add a detector (ionization chamber).

Open the machine's **Detector library** and add a chamber:

- Pick a model from the built-in catalog (e.g. an IBA FC65-G Farmer) or define your own
- Enter its **serial number** and **cavity volume / length**
- Enter the **$N_{D,w}$** calibration certificate — value, date, and validity period — and choose the **^{60}Co** or **cross-calibration** route

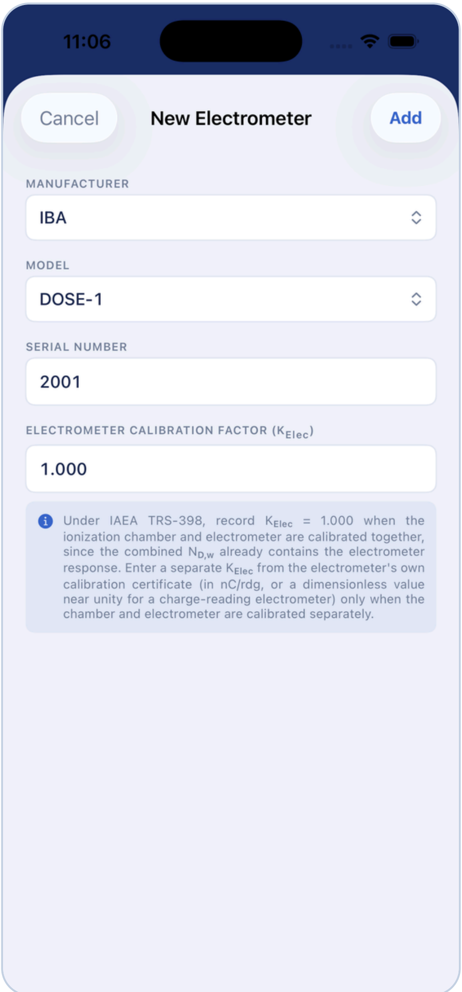
Save. Detector cards warn you as the certificate nears expiry.



New Detector — chamber, serial, and the $N_{D,w}$ calibration per electrometer.

4 Add an electrometer (optional).

In the **Electrometer library**, add make/model/serial and, if applicable, a calibration factor `k_elec` (default 1.000).



New Electrometer — make, model, serial, and k_{elec}.

5 Add a beam.

In the **Beam library**, add a **Photon** (or Electron) beam:

- **Energy** — e.g. 6 MV
- The **beam-quality index** your protocol needs — $TPR_{20,10}$, $\%dd(10)_x$, or R_{50}
- **Calibration conditions** — field size, delivered MU, and expected dose (used for the output check), plus PDD/TMR at the reference depth

Save.

New Photon Beam — energy, PDD_{10}, and $TPR_{20,10}$.

DEMO VALUES — REPRODUCE THE WORKED EXAMPLE

Enter these to reproduce the calculation shown in the next section, then adjust any of them to explore the app.

Machine	Varian TrueBeam · ID LINAC-1 · IAEA TRS-398 (2006) · SSD-based · reference depth 10 cm · 10×10 cm · 100 MU → 100 cGy expected
Electrometer	IBA DOSE-1 · S/N 2001 · k_{elec} 1.000
Detector	IBA FC65-G Farmer · S/N 1001 · 0.65 cm³ · cavity 2.31 cm · **$N_{D,w}$ 0.04776 Gy/nC** · ^{60}Co route
Photon beam	6 MV · **$TPR_{20,10}$ 0.670** · PDD_{10} 66.5% · d_{max} 1.5 cm
Conditions & reading	20.0 °C · 1013.3 hPa → k_{TP} 1.000 · **M = 14.02 nC** (single reading)
Correction factors	k_Q 0.9957 · k_{pol} 1.0010 (M_+ 10, M_- 10.02) · k_s 1.0030 (M_1 10.0000, M_2 9.9390, V_1/V_2 3, Pulsed) · k_{pw} 1.0000 · k_{elec} 1.0000
Result	**≈ 100.6 cGy at d_{max} (100.648) · +0.65 % vs 100 cGy expected**

Note: the app requires each detector and beam to be **locked** (Settings ▸ Detector / Beam Library) before it computes a dose, and the $k_{pol}/k_s/k_{pw}$ factors are set on the **Factors** tab. Any value can be changed at any time.

⚡ Shortcut — load a ready-made example

To skip the manual setup, use **Settings → Import Backup** and select a sample backup file: it loads a complete demo clinic (facility, machine, chamber, beam, and saved records) in one step. Export Backup does the reverse — everything to one shareable file.

03 Run a calculation

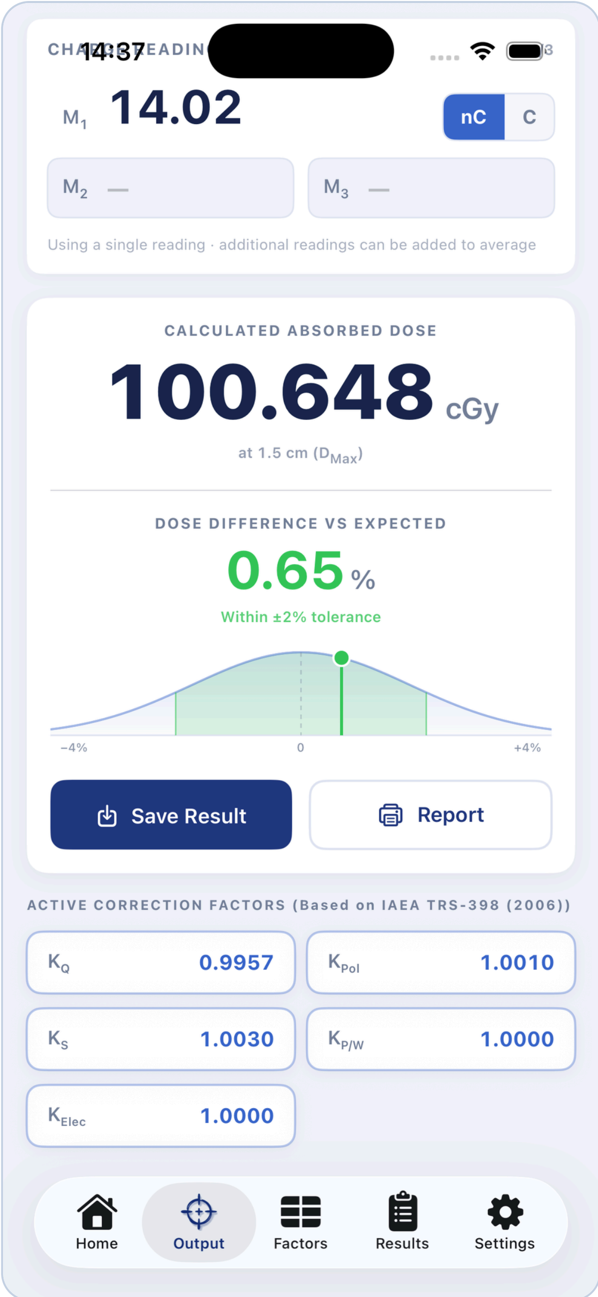
Open the **Output** tab. Your active facility, machine, detector and beam are shown at the top.

1. **Check the conditions.**
   Confirm **temperature** and **pressure** — the air-density correction k_{TP} is derived automatically.

2. **Select the detector and beam.**
   Tap your chamber under *Detector Information*, and your energy under *Beam Information* (Photon / Electron).

3. **Enter the charge reading.**
   Type M_1 (add M_2/M_3 to average up to three). The **absorbed dose to water** and the **% deviation vs expected** appear live on the gauge.

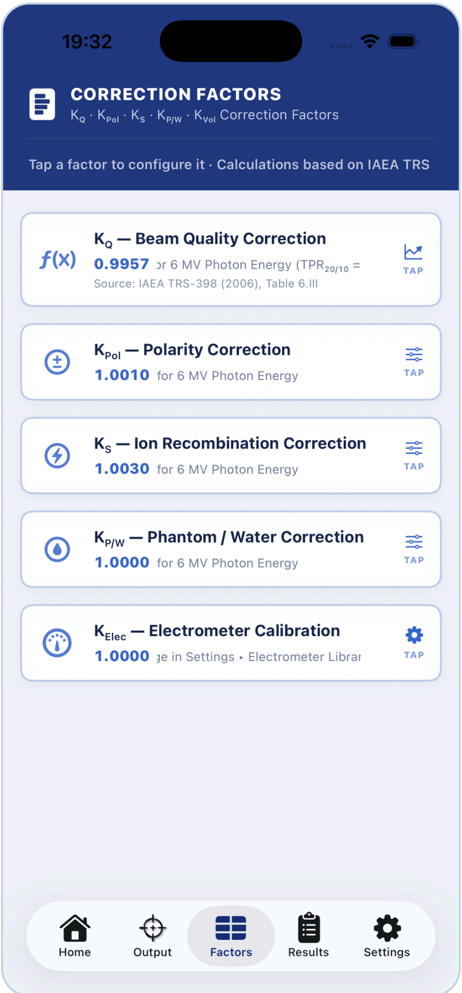
4. **Save or report.**
   Tap **Save Result** to record it, or **Report** to generate a PDF.



Output — the chamber reading becomes an absorbed dose (here 100.648 cGy, +0.65 % within tolerance), with the active correction factors listed below.

04 Correction factors

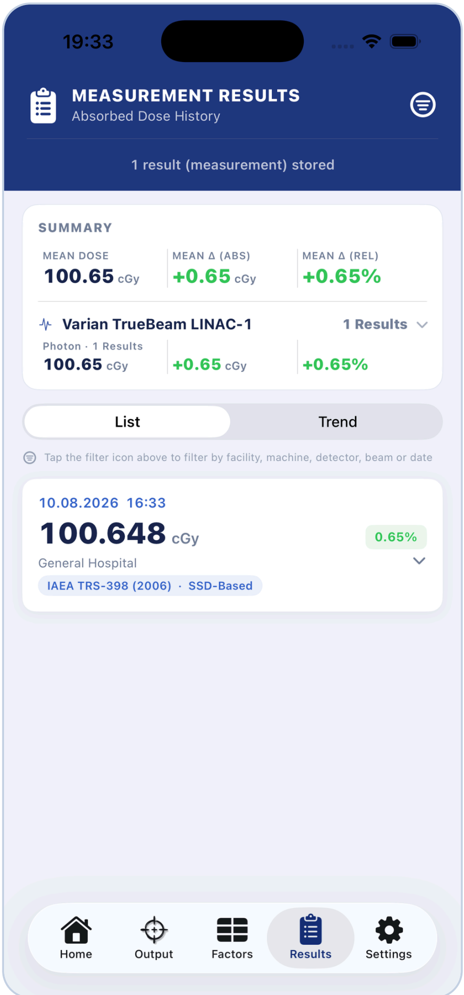
Open the **Factors** tab to measure and lock the corrections that aren't derived live — K_{Pol} (polarity), K_S (recombination), $K_{P/W}$ (phantom/water), and P_{gr} for TG-51 electrons. Each is stored **per chamber and energy**. A measured factor stays out of the dose until you **lock** it, so a half-entered value never silently changes a result.



Factors — every correction is shown and traceable; tap one to configure it.

05 Records & reporting

The **Results** tab keeps every saved calculation with its full factor snapshot — filterable by date and machine, with a scrolling trend chart so drift is visible at a glance. Use **Settings → Export Backup** to preserve or move your whole database.

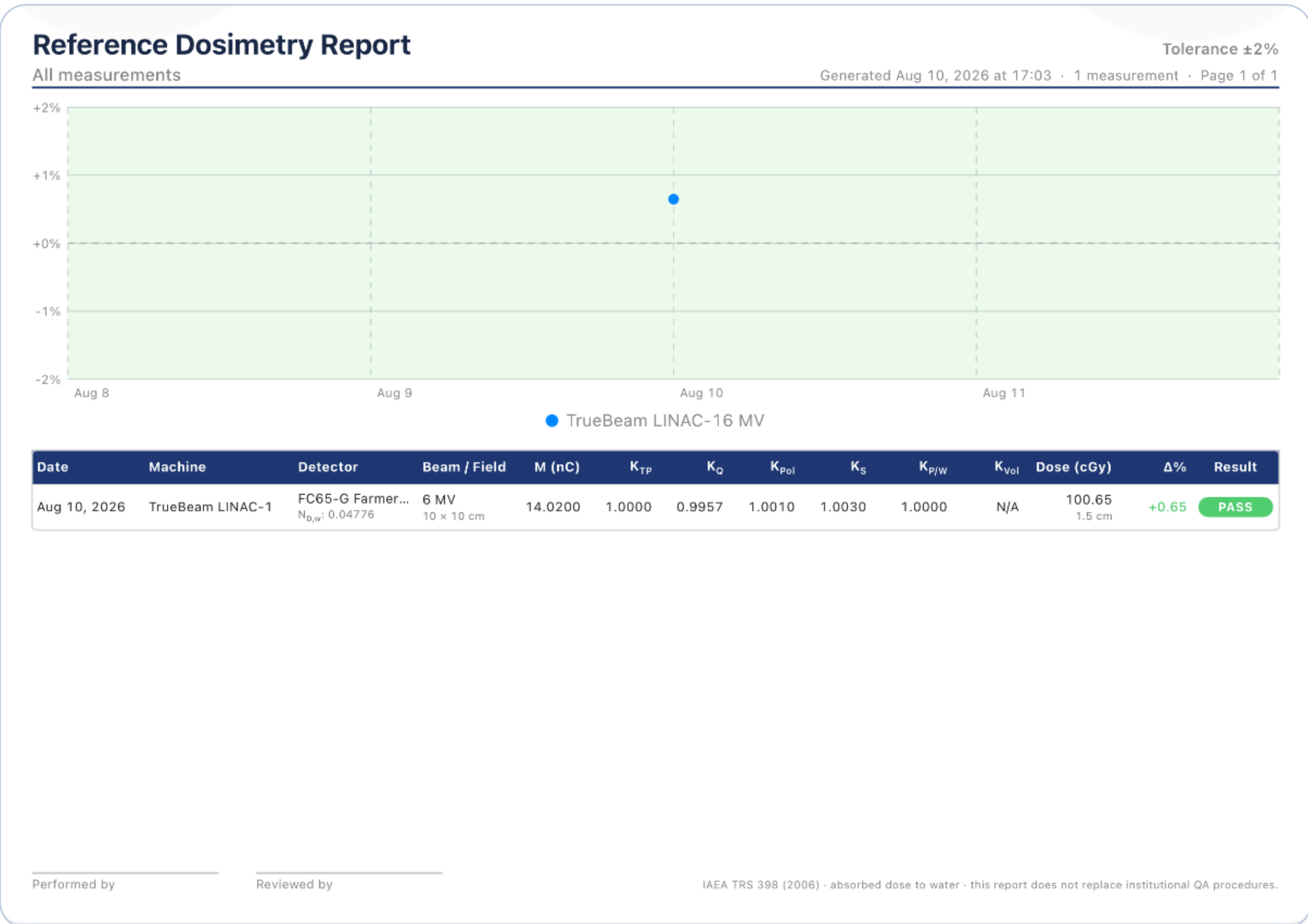


Results — the summary and every saved record; switch to Trend for the drift chart.

To generate a report, tap **Report** in the **Output** view (next to *Save Result*). Tick the records to include, then **Preview & Export**.

Reporting Records — select the measurements to include, then Preview & Export.

The result is a signed-off **A4 landscape PDF** — a trend chart above a full factor table (M, k_{TP}, k_Q, k_{Pol}, k_S, $k_{P/W}$, k_{Vol}), dose, deviation, and a Pass/Fail per record, with *Performed by | Reviewed by* sign-off lines — shareable or printable from the export button.



Date	Machine	Detector	Beam / Field	M (nC)	K_{TP}	K_Q	K_{Pol}	K_S	$K_{P/W}$	K_{Vol}	Dose (cGy)	Δ%	Result
Aug 10, 2026	TrueBeam LINAC-1	FC65-G Farmer… $N_{D,w}$: 0.04776	6 MV 10 × 10 cm	14.0200	1.0000	0.9957	1.0010	1.0030	1.0000	N/A	100.65 1.5 cm	+0.65	PASS

A4 · Landscape — the exported Reference Dosimetry Report (chart + full factor table, Pass/Fail per record).

Remember

RadiX Dose supports professionals — it never replaces the protocols, your judgement, or independent verification. Always verify every input, factor, and dose against the relevant code of practice before clinical use.

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